THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 2704	Use $(a \pm b)^2$
2	**(d)** $x^2 + 4x + 4$	Expand
3	**(c)** 64	$a^2 + 2ab + b^2$
4	**(c)** 420 km	60 km/h
5	**(a)** 180 minutes	Unit conversion
6	**(a)** 12 L	3/5 of 20
7	**(b)** 10%	Profit or loss on CP
8	**(a)** ₹ 1200	SP = MP − discount
9	**(d)** another Pythagorean triplet	Scaling
10	**(a)** 40 cm	$\sqrt{41^2 - 9^2} = 40$
11	**(c)** 4 km	20/5
12	**(a)** halves	Inverse proportion

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$x^2 + 12x + 36$ (2).	2
14	$75 : 300 = 1 : 4$ (2).	2
15	25% (2).	2
16	$100 + 576 = 676 = 26^2$, yes (2).	2
17	2 : 3 : 5 (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Write 304 as a sum or difference near a round number (1); apply $(a \pm b)^2$ (1); $304^2 = 92416$ (1).	1+1+1
19	Profit $= \frac{6}{100} \times 2500 =$ ₹150 (2); SP $=$ ₹2650 (1).	2+1
20	Diagonal 25 m (1.5); saved $31 - 25 = 6$ m (1.5).	1½+1½
21	Distance $= 340$ km (1.5); time $= 5.66667$ h (1.5).	1½+1½
22	By the theorem, height $= \sqrt{29^2 - 20^2}$ (2); $= 21$ m (1).	2+1
	OR: Triplet $(3, 4, 5)$ (1.5); $9 + 16 = 25 = 25$ (1.5).	1½+1½
23	Total parts $= 9$ (1); one part $=$ ₹200 (1); shares ₹ 400, ₹ 600, ₹ 800 (1).	1+1+1
	OR: 1 cm = 4 km (1); $1 : 400000$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 300 and 180 (2). (ii) 280 : 200 = 7 : 5 (2). (iii) Ratio changes as numbers shift (1).	2+2+1
	OR: (i) $36 \times 1000/3600 = 10$ m/s (2). (ii) 18 km/h (3).	2+3
25	(i) 120°, 90°, 60°, 45°, 45° (3). (ii) Sleep (1). (iii) 15° (1).	3+1+1
	OR: (i) 30 cm (2). (ii) 2.4 m (1). (iii) 1 : 22500 (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** 0	Zero property
2	**(b)** missing the term $2ab$	Mind the mistake
3	**(c)** 2 : 7	12 : 42
4	**(a)** 3 : 4	Divide both by HCF 6
5	**(a)** have the same value when simplified	Proportion
6	**(a)** twice the cost price	SP = 2 CP
7	**(b)** 15%	Profit or loss on CP
8	**(d)** 6.25%	Multiply by 100
9	**(a)** 13 cm	$\sqrt{5^2+12^2}=13$
10	**(d)** construct altars using rectangles and squares	Vedic geometry
11	**(d)** 12	Workers × days is constant
12	**(d)** 120 km	Multiply by the scale

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	No: $25 \neq 13$ (2).	2
14	Yes, $4 \times 27 = 108 = 9 \times 12$ (2).	2
15	₹ 900 (2).	2
16	In a right triangle, the square on the hypotenuse equals the sum of the squares on the other two sides (2).	2
17	20% (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$(a+b)(a-b) = (41+39)(41-39)$ (2); $= 160$ (1).	2+1
19	Let numbers be $7x$, $3x$; $10x = 50$ (1); $x = 5$ (1); numbers 35 and 15 (1).	1+1+1
20	Diagonal 5 m (1.5); saved $7 - 5 = 2$ m (1.5).	1½+1½
21	Total parts $= 11$ (1); one part $=$ ₹400 (1); shares ₹ 400, ₹ 1200, ₹ 2800 (1).	1+1+1
22	Distance $= 300$ km (1.5); time $= 5$ h (1.5).	1½+1½
	OR: Total work $= 360$ worker-days (1); days $= 20$ (1); difference 25 days (1).	1+1+1
23	Write 1002 as a sum or difference near a round number (1); apply $(a \pm b)^2$ (1); $1002^2 = 1004004$ (1).	1+1+1
	OR: $x^2 + 15x + 44$ (2); at $x = 1$: $(5)(12) = 60 = 1 + 15 + 44$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Expansion: $a^2 - ab + ab - b^2$ (2). (ii) 2496 and 9991 (3).	2+3
	OR: (i) A square of side $a+b$ split into a^2, two ab and b^2 (3). (ii) $(70+3)^2 = 4900 + 420 + 9 = 5329$ (2).	3+2
25	(i) $8000(1+r)^2 = 9680$, $r = 10\%$ (3). (ii) 10648 (2).	3+2
	OR: (i) $A = 20000 \times 1.331 = 26620$, CI ₹ 6620 (3). (ii) SI ₹ 6000, CI exceeds by ₹ 620 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** $(a+b)^2$	Identity
2	**(a)** 1530	$(100+2)\times 15$ using the distributive property
3	**(d)** ₹ 400	A gets 5/9 of 720
4	**(a)** 15	Multiply by 3
5	**(d)** 35%	Multiply by 100
6	**(d)** ₹ 600	$SI=\frac{PRT}{100}$
7	**(b)** 1320	1500 × 88/100
8	**(c)** 4 cm	$\sqrt{5^2-3^2}=4$
9	**(a)** hypotenuse	Opposite the right angle
10	**(c)** the sum of squares on the other two sides	$c^2=a^2+b^2$
11	**(b)** 144°	Angle = percent × 3.6
12	**(d)** 3/10	3 of 10 parts

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$3600-9=3591$ (2).	2
14	Comparison of two quantities by division (1); $2:3$ (1).	1+1
15	Discount ₹ 120, SP ₹ 680 (2).	2
16	5 m (2).	2
17	8 days (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$(a+b)(a-b)=(97+3)(97-3)$ (2); $=9400$ (1).	2+1
19	Let numbers be $3x$, $8x$; $11x=66$ (1); $x=6$ (1); numbers 18 and 48 (1).	1+1+1
20	After first discount: ₹ 1600 (1.5); final SP ₹ 1440 (1.5).	1½+1½
21	Distance $=310$ km (1.5); time $=5.16667$ h (1.5).	1½+1½
22	Write 99 as a sum or difference near a round number (1); apply $(a\pm b)^2$ (1); $99^2=9801$ (1).	1+1+1
	OR: The middle term $10x$ is missing (1.5); correct is $x^2+10x+25$ (1.5).	1½+1½
23	Total parts $=14$ (1); one part $=$ ₹200 (1); shares ₹ 600, ₹ 800, ₹ 1400 (1).	1+1+1
	OR: Milk 35 L, water 5 L (1); new water $=15$ L (1); add 10 L (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 3 : 2 : 1 (1). (ii) 750 g, 500 g, 250 g (2). (iii) 20 people (2).	1+2+2
	OR: (i) Ratio is a comparison by division; proportion is equality of two ratios (2). (ii) $x=9$ (1.5). (iii) $6\times 15=90=9\times 10$, yes (1.5).	2+1½+1½
25	(i) $5\sqrt{3}\approx 8.66$ cm (3). (ii) $25\sqrt{3}\approx 43.3$ cm² (2).	3+2
	OR: (i) $400+9801=10201=101^2$ (3). (ii) 990 square units (2).	3+2

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